

# Space exploration

The aerals transmitted radio signals to Earth.

The main body of Sputnik 1 was just 23 in (58cm) in diameter.

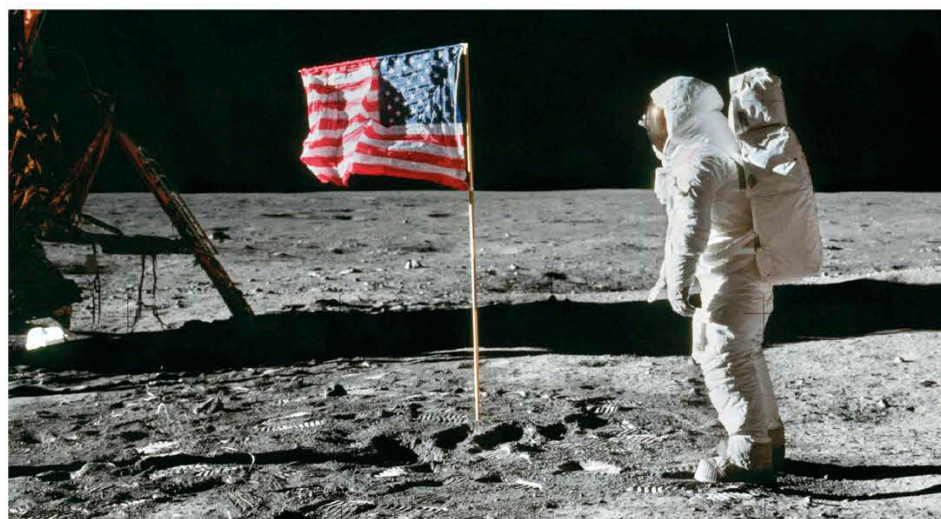
## ▲ SPUTNIK 1

On October 4, 1957, the USSR launched Sputnik 1, the first human-made object to orbit (travel around) Earth. It stayed in orbit for three weeks, and gathered valuable scientific data about Earth's atmosphere. Today, satellites are sent into space to collect information for weather forecasting, communications, spying, navigation, and other uses.

During World War II, scientists in Nazi Germany built V-2 rockets that were used as missiles. After the war, the Soviet Union (USSR) developed this technology to launch the world's first artificial satellite into space—*Sputnik 1*. This sparked a “space race” between the two world superpowers of the time, the USSR and the US, with each country trying to outdo

the other. While the Soviets were the first to send a human into space in 1961, the US won the race to land astronauts on the moon in 1969.

Since then, astronauts from around 40 countries have explored space, and many more missions have been launched without a crew. Some spacecraft have even left the solar system—the region of space in which Earth sits, along with the sun and seven other planets—to explore deep space and further our knowledge of the universe. Future crewed missions may even try sending humans to Mars!



## ◀ TO THE MOON

In 1961, after Yuri Gagarin's spaceflight, US president John F. Kennedy promised to land a human on the moon by the end of the decade. On July 20, 1969, two US astronauts—Neil Armstrong and Buzz Aldrin (left)—became the first humans to set foot on the moon's surface. Since then, other astronauts have walked on the moon, and many uncrewed space vehicles have orbited or landed on it.

## SPACE MILESTONES

**1944**

German scientists launch a V-2 rocket—the first human-made object to reach space.

**1958**

Cosmonaut (Russian astronaut) Yuri Gagarin becomes the first person in space, orbiting Earth for 1 hour and 48 minutes.

**1965**

Soviet cosmonaut Alexei Leonov performs the first “spacewalk”—getting out of a spacecraft into outer space.

**1981**

*Columbia*, the first reusable spacecraft, is launched by US space agency NASA. It goes on to fly 28 missions.